

Avanthika Shydh

Coimbatore | avanthika.shydh05@gmail.com | +91 9150646758 |

Summary

Quantitatively driven dual-degree engineering student pursuing a B.Tech in Electrical and Computer Engineering and a BS in Data Science & Applications from IIT Madras, with a strong foundation in machine learning and data science theory and practical experience in Python, C, SQL, and analytical problem-solving. Trained in scalable and enterprise-grade systems through coursework in Cloud Computing and Big Data Analytics, with a disciplined, research-oriented approach to building reliable and well-governed data solutions. Brings strong quantitative reasoning and the ability to communicate complex technical concepts clearly and concisely, enabling effective collaboration with cross-functional and non-technical stakeholders in a business-focused environment.

Education

Amrita School of Engineering Coimbatore, B.Tech in Electrical and Computer Engineering Aug 2023 – Present

- GPA: 8.20/10.0

IIT Madras , BS in Data Science and Applications

Sept 2023 - Present

- Diploma Level Ongoing

St.Antony's Public School, Anakkal

Mar 2021 – Mar 2023

- Score: 92.6%

SSVM Institutions ,Alangombu

Jun 2009 – Mar 2021

- Score:98.8%

Projects

Leveraging Retail Analytics to Optimize Staffing and Improve Sales in an Offline Clothing Store

- Delivered a real-world business analytics project for a retail company, gathering on-ground sales, customer footfall, and competitor pricing data to understand key operational and market challenges. Identified peak-hour staffing inefficiencies, gaps in pricing competitiveness, and a growing shift toward online buying behavior. Presented strategic, data-backed recommendations driving improved store performance and better customer experience.
- **Technologies:** Python, Pandas, Scikit-learn, Power BI, Excel

AI-Driven Traffic Monitoring System for Real-Time Congestion Detection and Route Optimization

- Engineered an AI-powered intelligent traffic system integrating YOLOv8 object detection and LiDAR fusion within CARLA simulation for real-time congestion prediction and adaptive signal control in 6G-enabled smart cities. Implemented proactive voice and SMS alert mechanisms, advancing safe, efficient, and autonomous urban mobility.
- **Presented this work as a research paper titled at ICISCoIS 2026, PSG College of Technology.**

Crime Type Prediction (Random Forest Classifier)

- Applied a Random Forest model achieving 95% accuracy in predicting crime types for Indian districts using 9000 rows of historical data with 29 features. Applied data preprocessing, feature engineering, hyperparameter tuning, and district-wise encoding. Evaluated performance using accuracy, precision, recall, and F1-score.
- **Technologies:** Python, Scikit-learn, Pandas, NumPy, Matplotlib

Face Authentication Enabled Hall Booking with Automated Power Control System

- Designed a smart college hall booking solution using face recognition for access validation, paired with microcontroller-enabled automated power control. Enhanced security, improved operational efficiency, and achieved significant energy savings during idle hours.
- **Technologies:** Raspberry Pi, Python (OpenCV), Flask, MySQL, HTML/CSS/JS, Relay-Based Power Automation

Skills

Programming: Python, SQL, C, MATLAB

Data: ETL, Data Cleaning, Data Modeling, Feature Engineering, Data Pipelines

Tools: Pandas, Power BI, Excel, NumPy, scikit-learn, Cognos

Domains: Data Management, Business Data Analytics, Retail Analytics, Smart Mobility Analytics

Certifications

IBM Data Analyst Professional Certificate